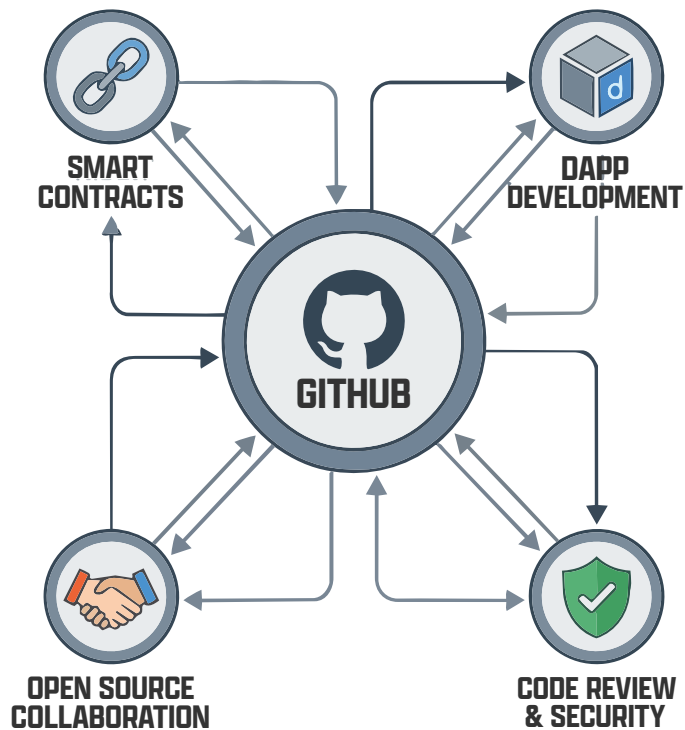


The Role of GitHub in Blockchain Development

GitHub is a key enabler of blockchain development as it formalises code collaboration and helps to transparently validate it before deployment. By embracing the principles of open source software, it allows contributors from around the world to develop, review and improve blockchain solutions.



GitHub is a foundational platform for all blockchain project management. It helps to track versions of code in a structured fashion, allows code to be shared openly, and facilitates collaboration by providing workflow tools that demonstrate the decentralised aspects of the blockchain. Developers use GitHub to develop smart contracts, manage their updates, maintain bug-tracking records, and coordinate their contributions among multiple geographic locations.

As a real-world example, developers use GitHub repositories to build and host a large part of Ethereum-based dApps. A developer creates a specific smart contract module (e.g. ERC20 Token), and other developers review it for possible vulnerabilities before it is integrated by a third party into a live application. This cycle of continual validation by other

developers ensures that the contribution is a safe piece of code for deployment on an unalterable blockchain.

Developers can use GitHub as a means of learning to evolve their careers. Many developers entering the Web3 ecosystem start by browsing the publicly available open source repositories, explore how others have built off those repositories, and then contribute their own smaller fixes or enhancements. Eventually, these contributions serve as a public portfolio, showing that the developer possesses relevant skills as well as experience working with others.

When attending a blockchain hackathon, students generally create clones of existing repositories and modify the logic of their smart contracts before contributing via pull requests to show the improvements made to each

project. This activity facilitates the operation of blockchain startups and decentralised development teams.

Web3 technologies like DeFi, NFT and DAO have strengthened the pivotal role of GitHub as a foundational resource for blockchain development. GitHub enables people from all around the world to collaborate on projects, creates transparency in the evolution of code, and facilitates scalable collaborative development in a decentralised environment. This has an impact on not only how applications are created, but also on how innovation is disseminated, validated and maintained in today's decentralised ecosystem. Hence GitHub can be viewed as more than a development platform -- it can be seen as a collaborative validation layer that confirms code reliability before it becomes part of an unalterable blockchain.